Training-source composition and wheat rust classification

A controlled study of training size and source diversity

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Abstract

A controlled 2 × 3 × 3 ablation studied the associations between fixed class-balanced training size (450 or 900 images), additional-source mixture fraction (0%, ~25%, or 50%), and wheat rust classifier performance across three random seeds. A MobileNetV2 transfer-learning classifier was evaluated on an earlier examined 489-image JIC benchmark and two protected image holdouts from source datasets represented in mixed-source training (IARI: 129; Ethiopia: 61). JIC mean accuracy was highest under 25% additional-source training (51.06% and 51.12% for 450/900 images), while the IARI and Ethiopian partitions generally favoured 50% (IARI: 95.09% at both training sizes; Ethiopia: 90.71%/96.17%). The contrast shows that observed performance rankings vary with evaluation distribution. Because the two additional-source partitions share source provenance with some training images, these results cannot establish adaptation to completely unseen domains.

1. Scientific question and experimental protocol

Question: with the number of images per disease class held constant, how do training-set size and source composition relate to accuracy and class discrimination on differently sourced evaluation images?

Parameter	Implemented setting
Classifier	MobileNetV2 with ImageNet pretrained frozen backbone; 224 × 224 RGB
Class order	Brown_rust / Healthy / Yellow_rust
Design	150 or 300 images per class × 0%, ~25%, 50% additional sources × seeds 42, 43, 44
Training recipe	Augmentation, Adam (learning rate 0.001), categorical cross-entropy; batch size 16; 10 epochs
Checkpoint	Best internal-validation loss; external outcomes not used to select checkpoint
Outputs	18 saved models; per-image evaluation outputs; condition means and sample SDs

The mixture is approximately 25% in the 450-image setting (114/450 = 25.33%) because class-level allocations are discrete. Original-source and mixed-source models use equal images per class but different source distributions.

2. Dataset provenance and safeguards

Partition or training pool	Image count	Coverage and interpretation
Primary train pool	2,942 initial	902 Brown / 1,116 Healthy / 924 Yellow; 1 screened out for later mixtures
Primary internal validation	737	Used for model-selection only
Staged IARI train	517	238 Brown rust mapping, 279 Healthy
Staged Ethiopia train	247	81 Healthy, 166 Yellow rust mapping
JIC benchmark	489	Other-source benchmark; previously evaluated, not newly blind
IARI original test	129	55 Brown, 74 Healthy; same dataset source represented in mixed training
Ethiopia holdout	61	20 Healthy, 41 Yellow; same dataset source represented in mixed training

The source-labelled IARI “diseased” category was mapped to Brown rust and Ethiopian “stripe_rust” to Yellow rust. These choices, especially the source-specific IARI disease mapping, require independent label/provenance confirmation before strong agronomic claims. IARI used its original test partition; Ethiopia used a seeded, image-level 20% split after duplicate-group screening.

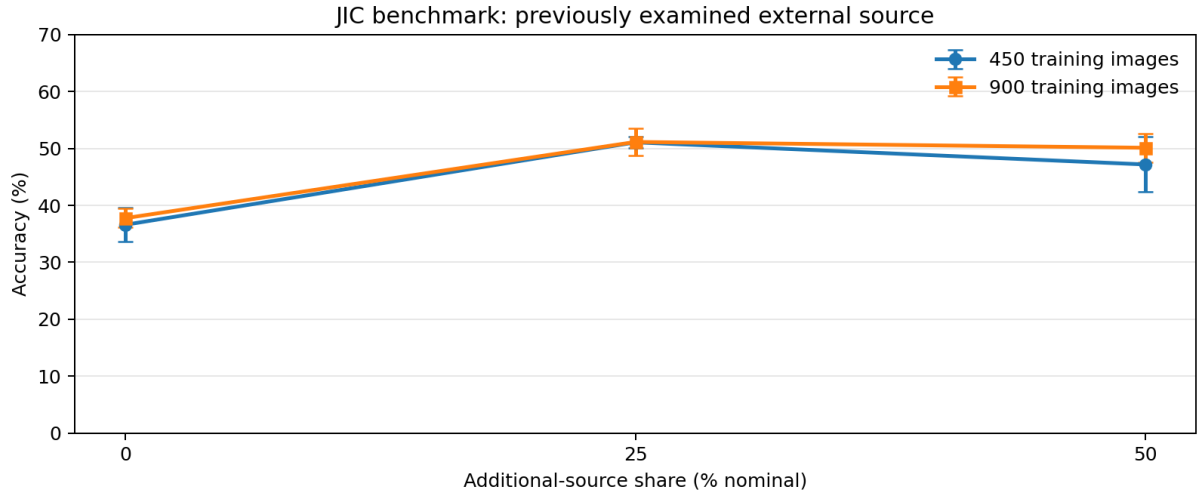
Image screening used exact SHA-256 and pHash distance ≤ 5. Logged exclusions include 2 Ethiopian similarity-group members, 63 IARI candidates flagged against test/validation, 21 training-similarity candidates, and 1 primary training candidate in mixture preparation. Perceptual similarities are conservative exclusion flags, not proof of same-plant or same-field duplicates. Image-level checks cannot establish site- or acquisition-session independence.

3. Main results: all conditions

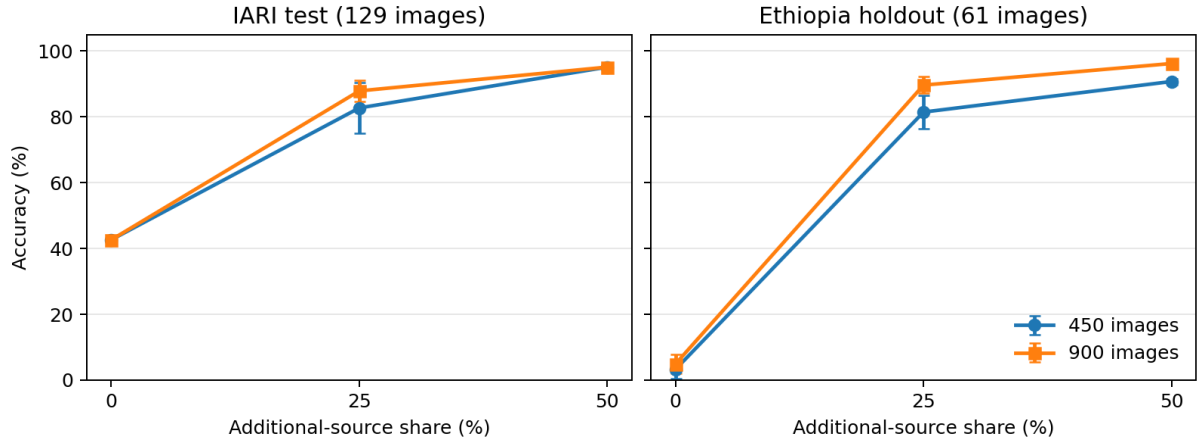
Each result is mean ± sample standard deviation in accuracy percentage points across seeds 42, 43, 44. These are descriptive seed SDs, not uncertainty intervals.

Train	Added source	JIC (489)	IARI (129)	Ethiopia (61)
450	0%	36.61 ± 2.95%	42.64 ± 0.00%	3.28 ± 2.84%
450	25%	51.06 ± 1.01%	82.69 ± 7.80%	81.42 ± 5.01%
450	50%	47.17 ± 4.84%	95.09 ± 1.18%	90.71 ± 0.95%
900	0%	37.76 ± 1.64%	42.64 ± 0.00%	4.92 ± 2.84%
900	25%	51.12 ± 2.41%	87.86 ± 3.23%	89.62 ± 2.50%
900	50%	50.10 ± 2.49%	95.09 ± 0.45%	96.17 ± 0.95%

JIC accuracy (means ± SD):



Reserved image holdouts within represented source datasets (means ± SD):



Complete per-run accuracy, macro-F1, balanced accuracy for present classes and absent-class prediction counts are included in the accompanying CSV. The dataset-specific class subsets prevent interpreting these as full three-class tests.

4. Confusion-matrix examination: matched seed-42 comparison

At training size 450 and random seed 42, the original-only model predicted Brown rust for all 129 IARI images, whereas the 50% added-source model recovered discrimination for 73/74 Healthy and 50/55 Brown rust leaves. This yielded 42.64% versus 95.35% IARI accuracy.

IARI actual class	Original-only correct	50% added-source correct
Brown rust (55)	55	50
Healthy (74)	0	73

On Ethiopian holdout, the original-only model predicted the absent Brown rust class for 57/61 images; the 50% model correctly classified 17/20 Healthy and 39/41 Yellow rust leaves. This yielded 6.56% versus 91.80% Ethiopian accuracy.

Ethiopia actual class	Original-only correct	50% added-source correct
Healthy (20)	4	17
Yellow rust (41)	0	39

This is direct evidence of class-prediction collapse in a particular original-only seed-42 model, followed by substantially better class discrimination in its paired mixed-source setting. It is not sufficient to diagnose a unique causal mechanism or certify generalisation to unseen field conditions.

5. Interpretation and explicit limitations

- On the previously examined JIC benchmark, 25% mixture had the highest three-seed mean accuracy for both training sizes; on source-matched IARI/Ethiopia partitions 50% performed best. No single mixture should be declared optimal across settings.
- IARI and Ethiopia supply both training and held-out images, so strong results there establish held-out-image performance on represented sources, not unseen-source adaptation.
- Additional-source identity and class coverage are coupled: IARI contributes Brown rust and Healthy, Ethiopia Yellow rust and Healthy. Differences in class mix, scene, background and label conventions may explain some changes.
- Test partitions include only two of three classes. Models can predict an absent third class; reported present-class macro-F1 differs from three-class macro-F1.
- The sample sizes (129/61 images, three random seeds), repeated JIC inspection and unknown plant/field grouping constrain generalisation and formal causal inference. Mean ± SD is descriptive only.
- Data-reuse permissions, original source labels, per-image provenance and possible same-plant/session leakage require third-party confirmation. Hash checks reduce one risk but do not close all pathways.

6. Reproducibility, audit and review questions

The accompanying reproducibility package contains the training, preparation, screening and evaluation scripts, complete per-run outcomes, condition-level summaries, chart data and a path-sanitized aggregate of both supplied manifests. Original photographs and large saved checkpoints are excluded. The physical run_metadata.json files and checkpoint SHA-256 files remain local to the researcher.

Evidence cross-check: 18 JIC records were transcribed from the PowerShell display of local run_metadata.json values, which passed the local 18-condition and image-count checks. The 18 holdout records and two stage/study manifests were provided as files. Selected confusion matrices were independently checked against pasted saved outputs. JIC source-value verification remains limited to the user-provided terminal output rather than copies of all 18 raw metadata files.

Suggested independent-review questions: (i) Do the comparisons adequately probe quantity versus source composition? (ii) What alternative explanations for the class collapse are plausible? (iii) Which dataset-provenance, label-mapping or independence issues require correction? (iv) What untouched independent-site evaluation would best test true out-of-domain generalisation?

Related public project: https://github.com/devanandha/wheat-rust-ai-detection

IARI source: https://data.mendeley.com/datasets/th422bg4yd/1 | Ethiopia source: https://data.mendeley.com/datasets/wgd66f8n6h/1